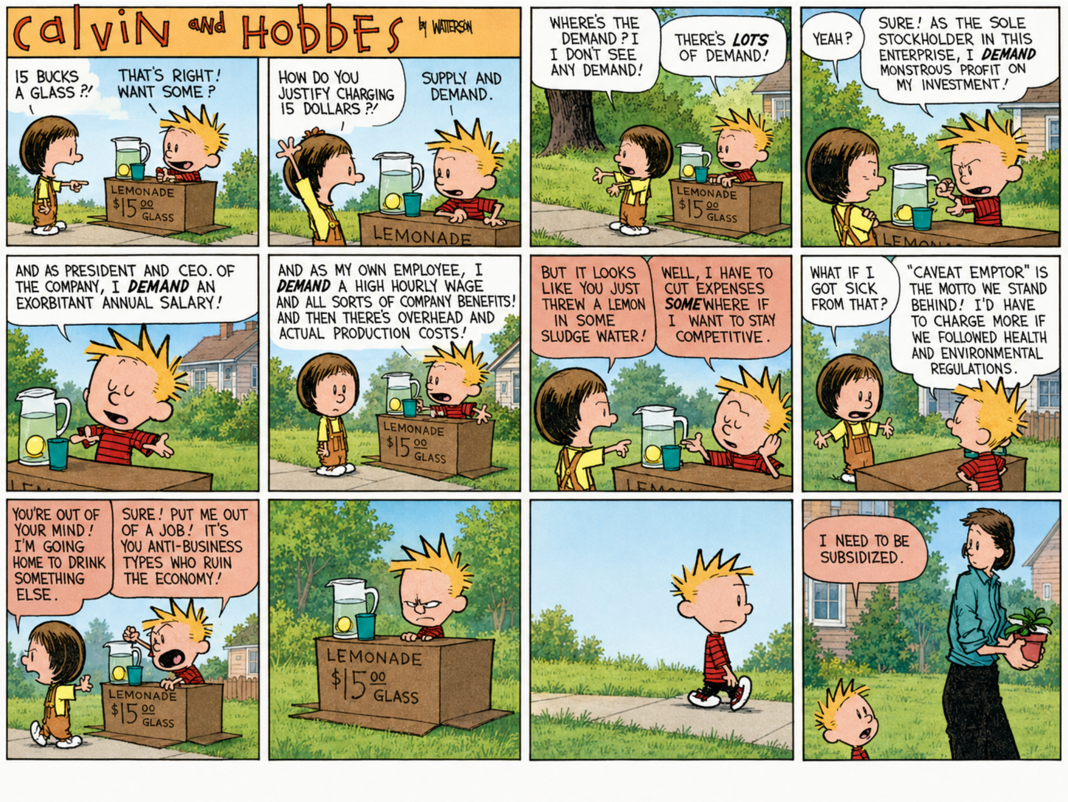
IITR Scaler Session - 9

We will begin 8:35 PM



Agenda

1. GAP -> Global Average Pooling
2. Convolution Network in Tensorflow

3 Transfer Learning

Global Average Pooling

224x224x3

10 classes

```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 conv2d (Conv2D)             (None, 224, 224, 32)      896

 conv2d_1 (Conv2D)           (None, 224, 224, 32)      9248

 max_pooling2d (MaxPooling2  (None, 112, 112, 32)      0
 D)

 conv2d_2 (Conv2D)           (None, 112, 112, 64)      18496

 conv2d_3 (Conv2D)           (None, 112, 112, 64)      36928

 max_pooling2d_1 (MaxPoolin  (None, 56, 56, 64)        0
 g2D)

 conv2d_4 (Conv2D)           (None, 56, 56, 128)       73856

 conv2d_5 (Conv2D)           (None, 56, 56, 128)       147584

 max_pooling2d_2 (MaxPoolin  (None, 28, 28, 128)       0
 g2D)

 conv2d_6 (Conv2D)           (None, 28, 28, 256)       295168

 conv2d_7 (Conv2D)           (None, 28, 28, 256)       590080

 max_pooling2d_3 (MaxPoolin  (None, 14, 14, 256)       0
 g2D)

 conv2d_8 (Conv2D)           (None, 14, 14, 512)       1180160

 conv2d_9 (Conv2D)           (None, 14, 14, 512)       2359808

 max_pooling2d_4 (MaxPoolin  (None, 7, 7, 512)         0
 g2D)

 flatten (Flatten)           (None, 25088)             0

 dense (Dense)               (None, 1024)              25691136

 dense_1 (Dense)             (None, 1024)              1049600

 dense_2 (Dense)             (None, 10)                10250

=================================================================
Total params: 31463210 (120.02 MB)
Trainable params: 31463210 (120.02 MB)
Non-trainable params: 0 (0.00 Byte)
_________________________________________________________________
```

224x224x3

Model: "sequential"

Layer (type)	Output Shape	Param #	
conv2d (Conv2D)	(None, 224, 224, 32)	896	224x224x3 0 (3x3x3)x32
conv2d_1 (Conv2D)	(None, 224, 224, 32)	9248	224x224x32 0 (3x3x32)x32
max_pooling2d (MaxPooling2D)	(None, 112, 112, 32) ↓m.p.	0	0 (3x3x32)x64
conv2d_2 (Conv2D)	(None, 112, 112, 64)	18496	
conv2d_3 (Conv2D)	(None, 112, 112, 64)	36928	
max_pooling2d_1 (MaxPooling2D)	(None, 56, 56, 64)	0	
conv2d_4 (Conv2D)	(None, 56, 56, 128)	73856	
conv2d_5 (Conv2D)	(None, 56, 56, 128)	147584	
max_pooling2d_2 (MaxPooling2D)	(None, 28, 28, 128)	0	
conv2d_6 (Conv2D)	(None, 28, 28, 256)	295168	
conv2d_7 (Conv2D)	(None, 28, 28, 256)	590080	
max_pooling2d_3 (MaxPooling2D)	(None, 14, 14, 256)	0	
conv2d_8 (Conv2D)	(None, 14, 14, 512)	1180160	
conv2d_9 (Conv2D)	(None, 14, 14, 512)	2359808	
max_pooling2d_4 (MaxPooling2D)	(None, 7, 7, 512)	0	7x7x512 →
flatten (Flatten)	(None, 25088)	0	= (7 * 7* 512) * 1024 + 1024
dense (Dense)	(None, 1024)	25691136	
dense_1 (Dense)	(None, 1024)	1049600	
dense_2 (Dense)	(None, 10)	10250	

Total params: 31463210 (120.02 MB)
Trainable params: 31463210 (120.02 MB)
Non-trainable params: 0 (0.00 Byte)

$7 \times 7 \times 512 \rightarrow$ Flatten $\rightarrow$ Dense (1024)

Prev-layer $\odot$ $(3 \times 3 \times \_) \times 512 \rightarrow 7 \times 7 \times 512$

\# of Kernel

Each of the channel contains information about some feature.

7 1 7 2 7 3 - - - - - 7 512

K1 K2 K3 K 512

Avg1 Avg2 Avg3 A

high avg. if 123 Found the Feature it was looking For.

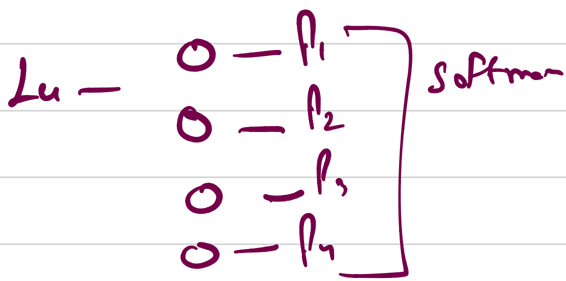
[Avg1, Avg2 ... Avg 512] $\longrightarrow$ Dense Layer

Credit Score Prediction → $-\infty$ to ∞

$F_1, \ldots, F_{10}$

$L_1 - 100 \rightarrow L_2 - 50 \rightarrow L_3 - 25 - L_4 - 1$

Sigmoid -> output to be b/w 0 and 1, when you want probability.
Relu -> I want to introduce non-linearity.
Linear -> For all continuous numerical value prediction
Softmax -> I want to have probability b/w 0 to 1 for multiple classes.

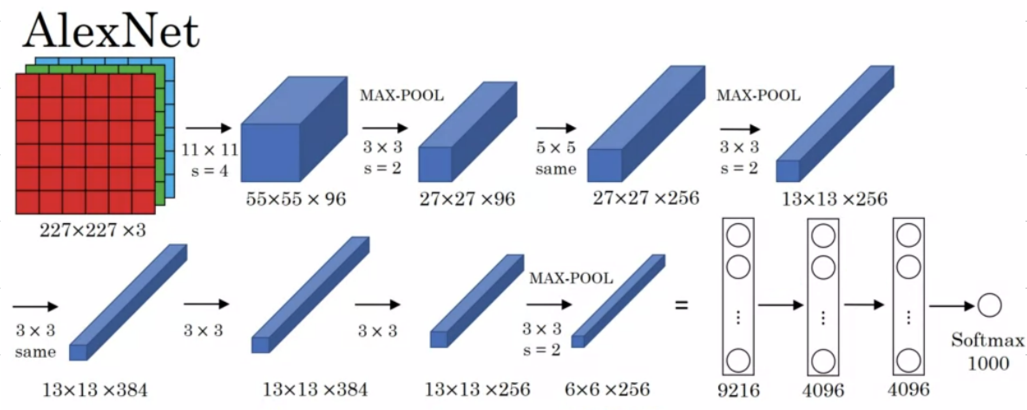


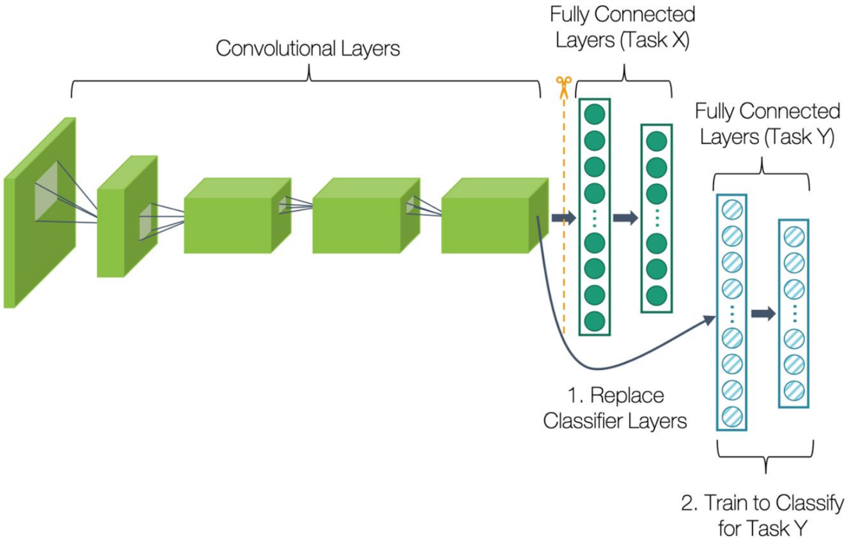
$p_1 + p_2 + p_3 + p_4 = 1$

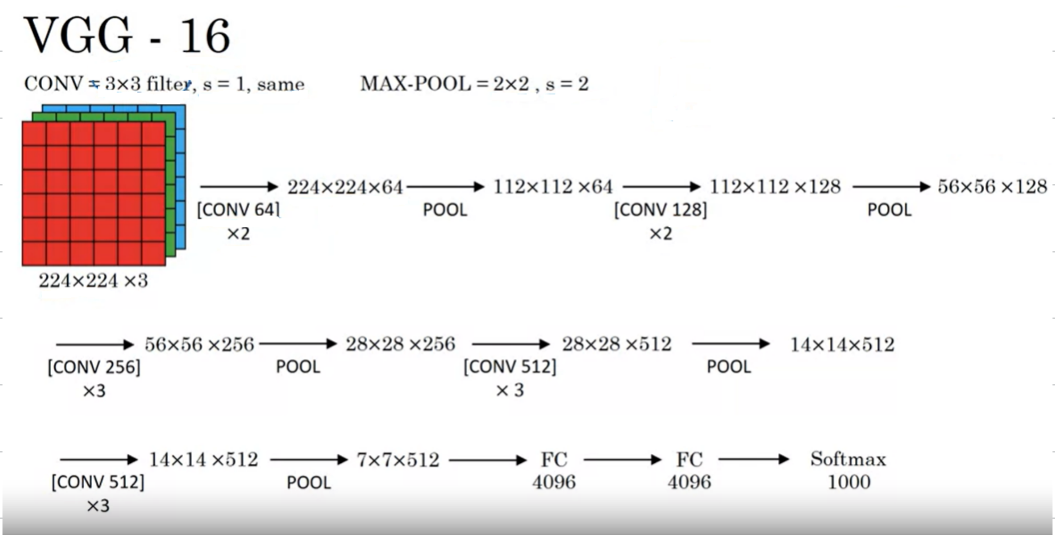
Sigmoid -> loan-approval or not, churn-probability prediction, spam or not, fraud or not

softmax -> predict among variety of fishes (tune, shark, hammer-head, blue-whale)
predict sentiment (extremely negative, negative, positive, extremely positive)

(kernel_height * kernel_width * input_channels + bias) * 96




Convolutional Layers
Fully Connected Layers (Task X)
Fully Connected Layers (Task Y)
1. Replace Classifier Layers
2. Train to Classify for Task Y


VGG - 16
CONV = 3×3 filter, s = 1, same
MAX-POOL = 2×2 , s = 2
224×224 ×3
[CONV 64] ×2
224×224×64
POOL
112×112 ×64
[CONV 128] ×2
112×112 ×128
POOL
56×56 ×128
[CONV 256] ×3
56×56 ×256
POOL
28×28 ×256
[CONV 512] × 3
28×28 ×512
POOL
14×14×512
[CONV 512] ×3
14×14 ×512
POOL
7×7×512
FC 4096
FC 4096
Softmax 1000